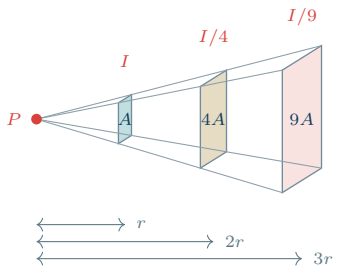
GEOMETRY AND THERMAL PHYSICS ON ONE PAGE

From $4\pi r^2$ to $\xi(2)$, $\xi(3)$ and $\xi(4)$

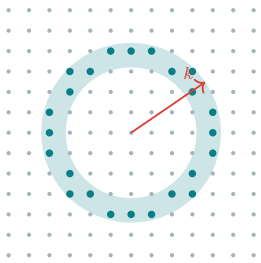
The sphere's area sets the inverse-square law and the photon mode count, and Bose sums give zeta values.

1 / THE RULE AND THE PICTURE

Area grows as r^{d-1}, so a conserved flux falls as $1/r^{d-1}$, which is $1/r^2$ in our space



same power over $A, 4A, 9A$



allowed wavevectors;
the shell holds $S_{d-1}k^{d-1}dk$ modes

A source of power P spreads over a sphere of area $S_{d-1}r^{d-1}$ in d dimensions, where $S_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is 2, 2π and 4π for $d = 1, 2, 3$. If nothing is absorbed the same power crosses every sphere, so a square of area A at distance r has a cone through it that cuts $4A$ at $2r$ and $9A$ at $3r$, and

$$I(r) = \frac{P}{S_{d-1}r^{d-1}} \xrightarrow{d=3} \frac{P}{4\pi r^2}$$

2 / TWO SUMS FROM THE INVERSE-SQUARE LAW

Shells of sources give a constant, and lamps at integer distances give $\zeta(2)$

A shell of thickness dr holds $4\pi r^2\,\rho\,dr$ sources of power P and each delivers $\frac{P}{4\pi r^2}$, so every shell adds $\rho P\,dr$ whatever its radius, and the total grows with the radius of the filled space, which is Olbers' paradox. Equal lamps at distances $1, 2, 3, \ldots$ on one side of a point, none blocking another, give instead $\sum_{n\geq 1}\frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6} = 1.645$ times the nearest lamp alone, which is 1.5498 after 10 lamps and 1.6350 after 100.

3 / THE SAME SPHERE COUNTS THE PHOTON MODES

In wavevector space the shell has area $S_{d-1}k^{d-1}$, and the Bose series turns it into zeta values

In a box the allowed wavevectors form a lattice, so per unit volume there are $\frac{g\,S_{d-1}k^{d-1}dk}{(2\pi)^d}$ modes between k and $k+dk$, with g the polarisation count ($g = 2$ and $\frac{k^2dk}{\pi^2}$ for light in $d = 3$). With $x = \frac{\hbar ck}{kT}$ and $\frac{1}{e^x-1} = \sum_{n\geq 1} e^{-nx}$ one has $\int_0^\infty \frac{x^{s-1}}{e^x-1}dx = \Gamma(s)\zeta(s)$, hence the energy density, while the number density has $\Gamma(d)\zeta(d)$ in place of $\Gamma(d+1)\zeta(d+1)$ and one power of kT fewer:

$$u_d = \frac{gS_{d-1}\Gamma(d+1)\zeta(d+1)}{(2\pi)^d}\,\frac{(kT)^{d+1}}{(\hbar c)^d}.$$

The nth term of the series is the gas at temperature T/n, whose wavevector volume is smaller by n^{-d}, so the number density carries $\sum n^{-d} = \zeta(d)$ and the energy density, with one more power of T, carries $\zeta(d+1)$.

4 / THE COMPLETED ZETA FUNCTION ABSORBS THE SPHERE

With $\xi(s) = \pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ the energy density is $u_d = g\,d\,\xi(d+1)\,\frac{(kT)^{d+1}}{(\hbar c)^d}$

The surface area and the gamma function combine exactly, $\frac{S_{d-1}\Gamma(d+1)\zeta(d+1)}{(2\pi)^d} = d\,\xi(d+1)$, which holds for every d. The table shows where each value appears.

dimension	energy density u_d	number density n_d	zeta values
$d=1$, $g=1$ (a line)	$u = \xi(2)\frac{(kT)^2}{\hbar c} = \frac{\pi}{6}\frac{(kT)^2}{\hbar c}$	diverges, since $\zeta(1)$ does	$\xi(2)$ in the energy
$d=2$, $g=2$ (a plane)	$u = 4\xi(3)\frac{(kT)^3}{(\hbar c)^2} = \frac{2\zeta(3)}{\pi}\frac{(kT)^3}{(\hbar c)^2}$	$n = \xi(2)\left(\frac{kT}{\hbar c}\right)^2$	$\xi(2)$ number, $\xi(3)$ energy
$d=3$, $g=2$ (our space)	$u = 6\xi(4)\frac{(kT)^4}{(\hbar c)^3} = \frac{\pi^2}{15}\frac{(kT)^4}{(\hbar c)^3}$	$n = \frac{4}{\pi}\xi(3)\left(\frac{kT}{\hbar c}\right)^3 = \frac{2\zeta(3)}{\pi^2}\left(\frac{kT}{\hbar c}\right)^3$	$\xi(3)$ number, $\xi(4)$ energy

So in our space the number of photons carries $\xi(3)$ and their energy $\xi(4)$, while $\xi(2) = \frac{\pi}{6}$ belongs one dimension lower, to the energy of a line and the number in a plane.

5 / NUMBERS IN THREE DIMENSIONS

The factor $\frac{1}{4}$ is the sphere again, and the cosmic background is a photon gas

The energy flux out of a hole in a box is $\frac{cu}{4} = \sigma T^4$ with $\sigma = \frac{\pi^2k^4}{60\hbar^3c^2} = 5.670\times10^{-8}$ W m^{-2}K^{-4}, and the $\frac{1}{4}$ is $\frac{\int\cos\theta\,d\Omega}{4\pi} = \frac{\pi}{4\pi}$, the same ratio of disc to sphere that makes the sunlight on the Earth average $\frac{S}{4}$: $\frac{0.7\times1361}{4} = 238$ W/m^2 balances σT^4 at $T = 254.6$ K. At $T = 2.7255$ K the background radiation has $n = \frac{2\zeta(3)}{\pi^2}(\frac{kT}{\hbar c})^3 = 411$ photons per cm^3 and $u = 0.2606$ eV/cm^3. In one dimension a single mode carries the power $\frac{\pi^2}{6}\frac{(kT)^2}{h} = 0.473$ pW at 1 K, whose rate $\frac{\pi^2k^2T}{3h} = 9.46\times10^{-13}$ W/K^2 times T is the quantum of thermal conductance.

ASIDE / TRY THESE

Check that $d = 3$ gives $6\xi(4) = \frac{\pi^2}{15}$ and that $\frac{2\zeta(3)}{\pi^2} = \frac{4}{\pi}\xi(3)$, and find the d for which the formula gives $\xi(5)$.

LINKS TO OTHER SHEETS

The dyadic sheet defines ξ and gives $\xi(2) = \frac{\pi}{6}$, the calculus sheet derives the factor $\frac{1}{4}$ for sunlight, and the Fourier sheet treats the wavevector lattice.

Convention. S_{d-1} area of the unit sphere in d dimensions; g polarisation count; $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$; all identities and numbers were checked to many digits.

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